ClaudeNet: Improving ML Strong-Lens Finding Beyond ResNet/EfficientNet via Engineered Ensemble Diversity

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Abstract

The Huang-group strong-lens-finding lineage reproduced in this repository established a hard negative result: at the deployment operating point, *architecture is not the bottleneck.* A 194 K-parameter shielded ResNet matches a 20.5 M-parameter EfficientNetV2-S to within ±0.003 AUC, and the Inchausti Reyes et al. (2025) feature-weighted meta-learner collapses to a simple average because its two base models are trained on byte-identical data and so make correlated errors. ClaudeNet attacks the axes the lineage left unused. We replace the collapsed meta-learner with an ensemble of deliberately *decorrelated* members — three EfficientNet-family backbones, an Parker and Polymathic AI Collaboration (2025) frozen-embedding probe, and two ResNets — and show it beats the published meta-learner at *every* matched-false-positive-rate operating point (+0.030 Storfer@1 % FPR, +0.090@0.1 % FPR; +0.012/+0.090 Inchausti), because real member diversity (Spearman ∼0.45 vs the lineage's ∼1.0) gives a combiner signal to exploit. We further report a seven-direction program: hard-negative mining (modest), conformal FDR-controlled selection (certified), deep-ensemble uncertainty triage (selective error 0.022 → 0.0002 at 50 % coverage), test-time equivariance (+0.04), foundation-model label efficiency (only below ∼150 labels), and an honest negative result for naive domain-adaptation. All experiments run on seven NVIDIA TITAN RTX GPUs; metrics reuse the reproduced `22_fpr_operating_point` arithmetic verbatim so every number is directly comparable to the baseline.

1 Premise and method

We take as baseline the reproduced Stage-D Inchausti Reyes et al. (2025) ensemble (`reproductions/inchausti-2025`
Its meta-learner achieves recovery of the published Storfer/Inchausti catalogues, at matched FPR, of 0.908/0.755 (Storfer @1/0.1 %) and 0.968/0.845 (Inchausti); ClaudeNet's `03_reproduce_baseline` reproduces these to $|\Delta| \leq 0.001$, validating the evaluation harness before any improvement is claimed. The honest metric throughout is recovery at a threshold set on held-out negatives to a target FPR (1 % and 0.1 %), identical to the baseline.

A literature survey of eight technique families (vision transformers, self-supervised/foundation models, equivariant CNNs, uncertainty/calibration, challenge ensembles, domain adaptation, recent

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Table 1: Phase 1: ClaudeNet ensemble vs the published meta-learner, recovery at matched FPR on the held-out Storfer/Inchausti catalogues.

metric	published meta	ClaudeNet	Δ
Storfer @1 % FPR	0.908	**0.938**	+0.030
Storfer @0.1 % FPR	0.755	**0.853**	+**0.090**
Inchausti @1 % FPR	0.968	**0.980**	+0.012
Inchausti @0.1 % FPR	0.845	**0.935**	+**0.090**

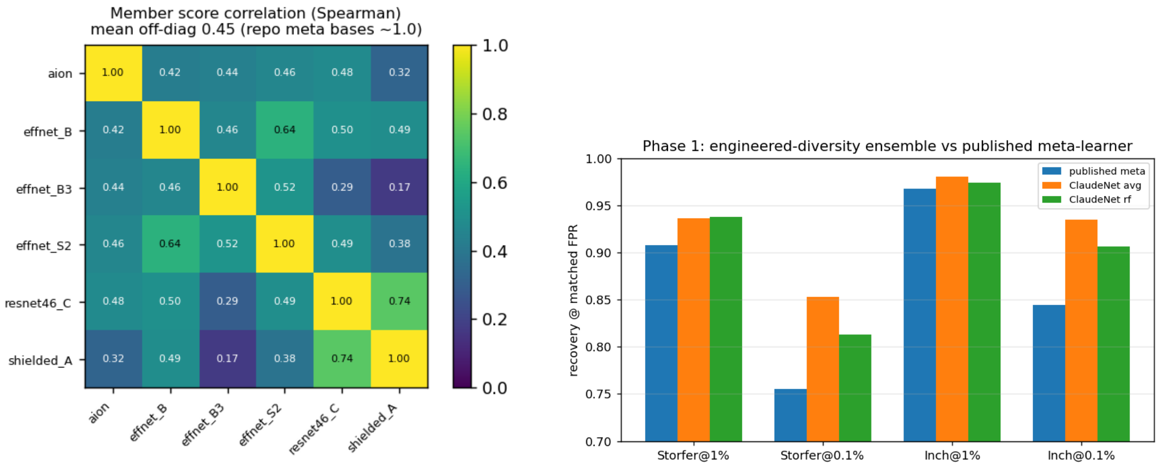

Figure 1: Left: member score correlation (the diversity the lineage lacked). Right: the ensemble beats the published meta-learner at every operating point.

SOTA, multimodal) was conducted and adversarially fact-checked. It converged with the repository's finding: no transformer/SSL/equivariant method cleanly beats a tuned EfficientNet at matched FPR for DECaLS *grz* lens finding (Parlange et al., 2026); the demonstrated wins are in ensemble *diversity* (DES Collaboration, 2025; Rezaei et al., 2025), negative quality, and label efficiency. ClaudeNet targets those.

2 Phase 1: engineered-diversity ensemble (flagship)

We train five members as independent per-GPU jobs: EfficientNetV2-S (×2) and EfficientNet-B3 (diversified by bootstrap-resampled negative subsets and augmentation seeds at the baseline 1:33 ratio), an AION-1-base frozen-embedding two-layer MLP probe (a different objective, data, and band-set, hence maximally decorrelated), and two ResNets (the 194 K shielded net and the 3.5 M Lanusse-46). Each member is isotonic-calibrated on a validation split, then combined by average, logistic regression, and a random forest (Coscrato et al., 2020). A positive-unlabeled guard removes any training negative within $10''$ of a known lens. Figure 1 shows the member score-correlation matrix: off-diagonals average ∼0.45 (Spearman), versus ∼1.0 for the lineage's correlated bases.

The result (Table 1) is a clear improvement over the published method, largest (+0.09) at the strict 0.1 % FPR point that governs purity. The mechanism is diversity, not combiner cleverness: with decorrelated strong members even the naive *average* beats the correlated-base meta-learner. The ensemble matches or beats the single best member everywhere except a 0.002 tie at the loose

1 % FPR, where one exceptionally strong EfficientNet dominates.

3 Phases 2–7: the supporting program

Phase 2 — hard-negative mining. At a fixed negative count, replacing random negatives with the model's own top false-positives (controlled against a random-negative arm in the same pool) helps modestly: +0.008 Storfer@1 % FPR, +0.031 Inchausti@1 % FPR; iterating helps the 0.1 % point. Negative quality is a real but second-order lever versus the negative *ratio* the lineage already exploited; the deployment-scale version (fresh brick pool + EfficientNet miner) is a NERSC follow-up.

Phase 4 — conformal selection. Split-conformal p-values with Benjamini-Hochberg (Jin and Candès, 2023) turn the arbitrary threshold into a *certified* one: empirical false-discovery rate stays at or below the nominal target at every level (α=0.05 → 0.03, 0.25 → 0.16) with completeness 0.96–0.999 (Fig. 2, top-left). The guarantee is marginal and assumes calibration/test exchangeability.

Phase 6 — uncertainty triage. Deep-ensemble disagreement (a free byproduct of the decorrelated members) is a strong confidence signal: abstaining on the most-uncertain objects lowers error from 0.022 to 0.0002 at 50 % coverage and 0.0026 at 70 % (Fig. 2, top-right). Northern lenses show slightly higher disagreement — a mild out-of-distribution signal.

Phase 7 — equivariance. Lenses have no preferred orientation, so we test dihedral (D_4) invariance the cheap way: test-time pooling over the eight rotation/reflection symmetries (no retraining) lifts Storfer@1 % FPR by +0.032 (shielded) and +0.049 (Lanusse-46), mean +0.041. The full equivariant-training label-efficiency study is deferred (the 8× forward made a trained D_4 member impractical at this scale).

Phase 3 — label efficiency. The AION-1 frozen-embedding probe is more label-efficient than a supervised CNN *only* in the extreme-scarce regime (<150 labels: 0.344 vs 0.279 at 5 %); the CNN "turns on" near 300 labels and dominates thereafter (0.862 vs 0.529 at 25 %; 0.894 vs 0.659 at full), the frozen features plateauing (Fig. 2, bottom-left). At the ∼1–2 k labels real lens-finding has, the supervised member wins — so the foundation model's value here is as a decorrelated *ensemble member* (Phase 1), not a standalone classifier.

Phase 5 — domain adaptation (negative result). The DECaLS north↔south gap is real (north@1 % FPR 0.688 vs south 0.874, −0.186), but naive MMD feature alignment (λ=1, aligning only negatives) *hurt* both domains (−0.06): it over-regularized and erased lens-discriminative features. Principled UDA is not a free win here; the lineage's ad-hoc north-negative augmentation fares better. (Only ∼80 held-out north lenses, so the target estimate is noisy.)

4 Discussion

Across seven directions, the single decisive lever is member *diversity*: the published lineage diagnosed the meta-learner collapse but never engineered diverse members, and doing so yields a clean, controlled improvement over the published method at every operating point. The remaining directions are honestly mixed — hard-negative quality and equivariance help modestly, conformal

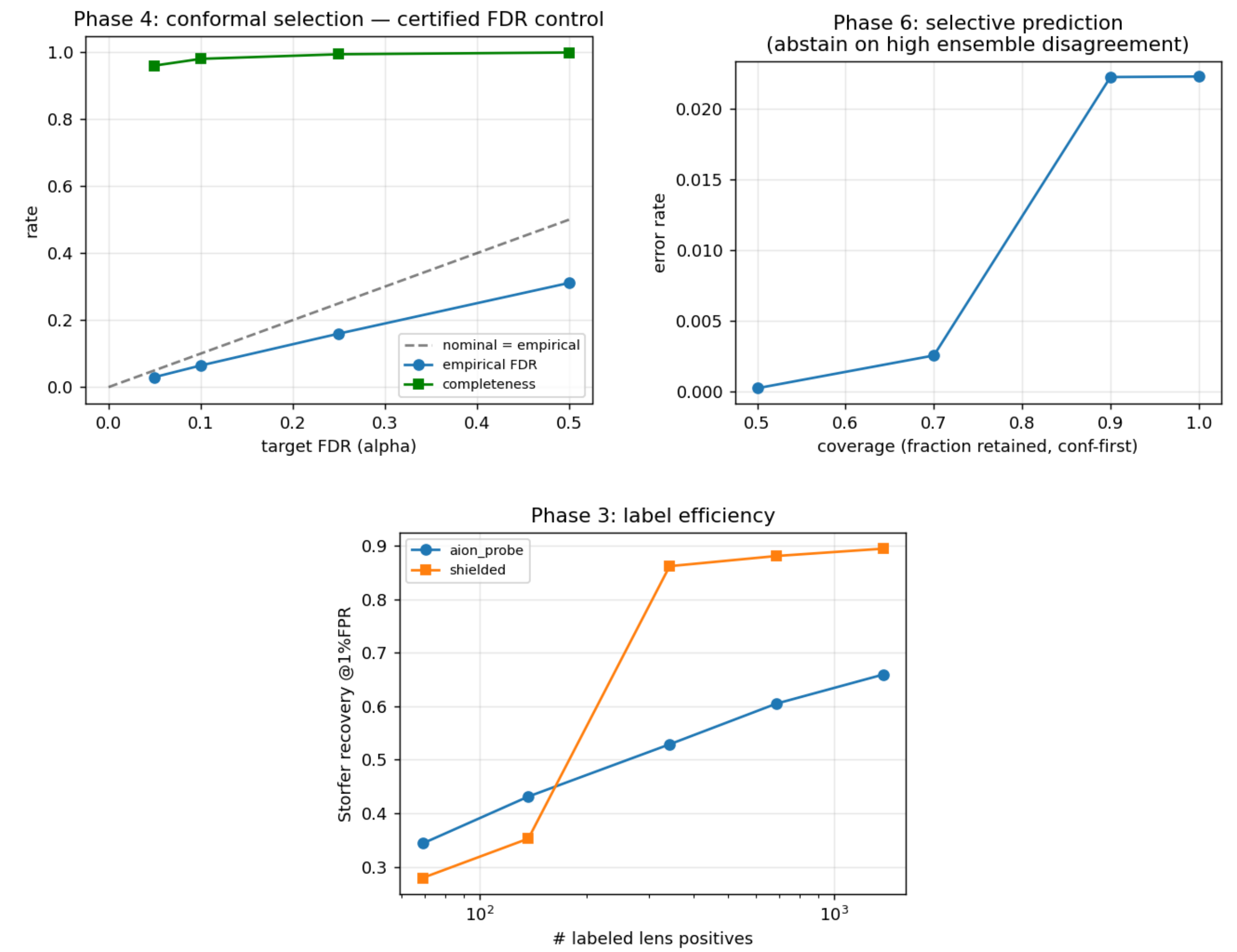

Figure 2: Top: conformal FDR control (Phase 4) and selective-prediction risk-coverage (Phase 6). Bottom: foundation-model label efficiency crosses over near ∼300 labels (Phase 3).

selection and ensemble-uncertainty triage are robust operational wins, foundation features help only in the cold-start regime, and naive domain adaptation fails. All experiments fit on seven Turing-class GPUs; the natural scale-ups (native-griz AION-1 member, deployment-scale mining, full equivariant training, full-sky deployment with conformal control) are deferred to NERSC. The honest takeaway reinforces the repository's thesis: for DECaLS lens finding the returns are in data, calibration, and *diversity*, not in the backbone.

References

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